

Hydrogeological and Hydroecological Assessment of Springs in Azerbaijan: Spatial Variability, Water Quality, and Sustainable Management

Rashail İsmayilov

Abstract

This study explores the hydrogeological characteristics of springs in Azerbaijan, focusing on their distribution, water quality, and potential for sustainable use. The country's diverse geological formations, including the Greater Caucasus, Lesser Caucasus, Talish mountain-fold zones, and the Kura-Araz Lowland, significantly influence the quantity and quality of groundwater. Groundwater in Azerbaijan is primarily found in fracture-porous and fracture-vein aquifers, with spring flow patterns following radial distribution from the center of the mountainous zones to their peripheries. Hydrogeological factors such as lithology, tectonic structures, and erosion zones play a critical role in determining spring formation, discharge rates, and mineralization levels. The study categorizes springs based on discharge, mineralization, temperature, and flow rate variability, with many springs exhibiting low to medium flow rates and mineralization that meets drinking water standards. Hydrogeological investigations confirm that most springs comply with national water quality standards, although areas with higher mineralization require careful monitoring. The research underscores the importance of sustainable water management practices to preserve spring-fed water resources, ensuring their long-term viability for local communities and ecosystems. The findings contribute to a comprehensive understanding of Azerbaijan's groundwater resources, highlighting the need for continued research and monitoring to safeguard these critical water supplies.

Keywords:

Hydrogeology, Hydro-ecology, Springs, Spatial Variability, Water Quality, Sustainable Management

Azerbaijan State Water Resources Agency "Water and Amelioration Scientific Institute" public legal entity, Baku Engineering University, Baku, Azerbaijan
e-mail: rashail.ismayilov@gmail.com

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1. Introduction

The Republic of Azerbaijan is located in the southeastern Caucasus, in the northwest of the Iranian Plateau, at the junction of the European and Asian continents. Its geographic location is highly favorable both in terms of natural conditions and geopolitical considerations. From a natural-geographical perspective, Azerbaijan's advantageous position is primarily determined by its location in the subtropical zone (at the transitional zone between the subtropical and temperate zones), the abundance of thermal resources, large plains with fertile soils, extensive foothills, and the presence of mountain systems that are rich in natural resources and have a diverse environment (İsmayilov *et al.*, 2026).

The Republic of Azerbaijan is characterized by diverse geographic and climatic conditions. This diversity plays a significant role in the water resources available to the country, particularly springs, which are a primary source of freshwater. In Republic of Azerbaijan, where agriculture and drinking water depend on consistent water availability, spring runoff is particularly important. It ensures the stability of water levels in key water bodies, mitigates drought risks, and supports irrigation systems. Without a reliable spring runoff, Republic of Azerbaijan

could face water scarcity, affecting its economy and population. Therefore, managing spring water resources is essential for the country's long-term water security (Geology of Azerbaijan, 2008; Alakbarov, 2014).

In recent years, due to the impact of anthropogenic factors on the environment, significant changes have occurred in the quantity and quality indicators of water sources, especially spring water, in the region. Currently, the implementation of various economic measures in the Republic of Azerbaijan, along with the effects of global climate change, has led to changes in both the quantity and quality indicators of spring water. Unfortunately, the changes in both the quantity and quality indicators of springs, which are one of the main sources of drinking water for the population of Azerbaijan, have not been sufficiently studied. In this regard, it is considered crucial to scientifically investigate the hydrogeological condition of the springs located in the area.

2. New approach to risk forecasting

The research conducted with support from the Azerbaijan State Water Resources Agency's "Water and Amelioration Scientific Research Institute" played a crucial role in evaluating the hydrogeological security of springs in

Azerbaijan. The use of archival materials, along with the author's own research, provided an in-depth understanding of the water resources of the springs, enhancing the overall analysis.

To properly evaluate the hydrogeological security of these springs, a multifaceted approach is required. This involves the integration of hydrological, geological, and ecological techniques to gain a comprehensive understanding of the springs' behavior, sustainability, and potential vulnerabilities.

Hydrogeological methods are essential for comprehensively understanding groundwater systems and the behavior of springs. These methods are rooted in the analysis of regional geology, which plays a crucial role in shaping how groundwater is stored, flows, and emerges at the surface as springs. These methods examine the properties of rocks and soils that affect how water is stored and flows underground, thereby influencing the overall functioning of the springs (İsmayilov & Kreame, 2026).

Ecological methods assess the impact of human activities, climate change, and environmental factors on the ecosystems surrounding the springs. These methods study how changes in water availability and quality affect plant and animal life, as well as the broader ecosystem. Additionally, ecological methods investigate the effects of pollution, land use changes, and climate variability on the spring ecosystems (İsmayilov *et al.*, 2026).

By combining these techniques, the study provides a holistic approach to understanding the hydrogeoeological characteristics of the springs. This integrated methodology helps assess the long-term sustainability and resilience of these vital water sources, ensuring their protection against both natural and human-induced changes. This approach contributes significantly to the sustainable management of water resources and the preservation of spring ecosystems in Azerbaijan (İsmayilov & Kreame, 2026).

The Great Caucasus, Lesser Caucasus, and Talish mountain-fold zones, primarily composed of Mesozoic and Cenozoic-aged rocks, are characterized by a sharply fragmented relief, a thick erosion zone, rock fissility, a low-thickness deluvial clay cover, numerous river valleys filled with alluvial and fluvio-glacial deposits, and intermountain depressions. In the mountainous region, the distribution of groundwater is not subject to any stratigraphic division but is mainly associated with erosion zones and tectonic disturbances. The water content of various aged rocks depends on their lithological composition, fissility, and the area's drainage conditions. The distribution pattern of spring flow in the mountainous areas shows that, similar to surface runoff, the subsurface flow is directed radially from the center of these zones to their peripheries. The geological and hydrogeological characteristics of these areas create favorable conditions for the manifestation of groundwater in the form of descending and ascending springs (Alakbarov, 2000; Geology of Azerbaijan, 2015).

The hydrogeological conditions of Azerbaijan have developed under the influence of a wide array of natural and anthropogenic factors, resulting in a complex and diverse landscape. The country's geological framework consists of the Lesser Caucasus, which is primarily made up of intrusive rocks; the Greater Caucasus, predominantly composed of sedimentary rocks; and the Kur-Araz

Lowland, situated between these two mountain ranges, which is characterized by both continental and marine sedimentary deposits (İsmayilov, 2019; İmanov & Alakbarov, 2017).

3. Engage experts

In the Republic of Azerbaijan, springs are formed through the natural emergence of groundwater onto the Earth's surface. The utilization of these springs as a water source is determined after conducting thorough hydrogeological investigations. These studies assess whether the water quality meets the existing standards for drinking water and whether the yield of the springs can meet the required volume for consumption. The hydrogeological characteristics of the region play a crucial role in both the quantity and quality of the water that the springs provide. Geological and hydrogeological factors, such as the presence of erosion zones, fractures, and deep tectonic faults, significantly influence the formation of these springs. These factors lead to the creation of fracture-porous and fracture-vein aquifers, which are essential in the movement and storage of groundwater. The characteristics of the groundwater, including its chemical composition and flow rates, depend largely on the geological structure of the region. Furthermore, the geological formation can determine whether the spring water is suitable for use without extensive treatment or if it requires purification processes. For a spring to be deemed reliable as a drinking water source, it must demonstrate consistent quality and sufficient yield, which are evaluated through detailed hydrogeological analysis. The balance between groundwater recharge and the extraction rate is essential for maintaining the sustainability of spring water sources in the region. Therefore, hydrogeological investigations are key to ensuring that spring water is both safe and sufficient for the population's needs (Barber & Wheeler, 2005).

According to geological-structural features, the territory of the Republic of Azerbaijan is divided into the Greater Caucasus and Lesser Caucasus fold zones, and the Kura-Araz Lowland between them. Although the Lesser Caucasus is separated from the Greater Caucasus by the Araz River in the southeast, this fold zone is also included in the geostructural zoning of the Lesser Caucasus. Based on the similarity of hydrogeological conditions, the analogous formation conditions of groundwater reserves, and the distribution patterns of hydrodynamic and hydrochemical parameters, 18 types of hydrogeological basins, including pore layers, pore-fault, and fault types, are identified within the boundaries of geostructural regions (Figure 1). Groundwater is a natural discharge of water directly to the Earth's surface or into water bodies (subsurface springs). A spring or a group of springs with high flow rates may form the source of a river under favorable natural conditions. Based on hydrodynamic indicators, springs can be descending or ascending. Descending springs are formed as a result of pressureless groundwater rising to the surface. These springs can be found in depressions such as river valleys, gorges, ravines, etc. Ascending springs form when pressurized groundwater reaches the surface (İmanov & Alakbarov, 2017).

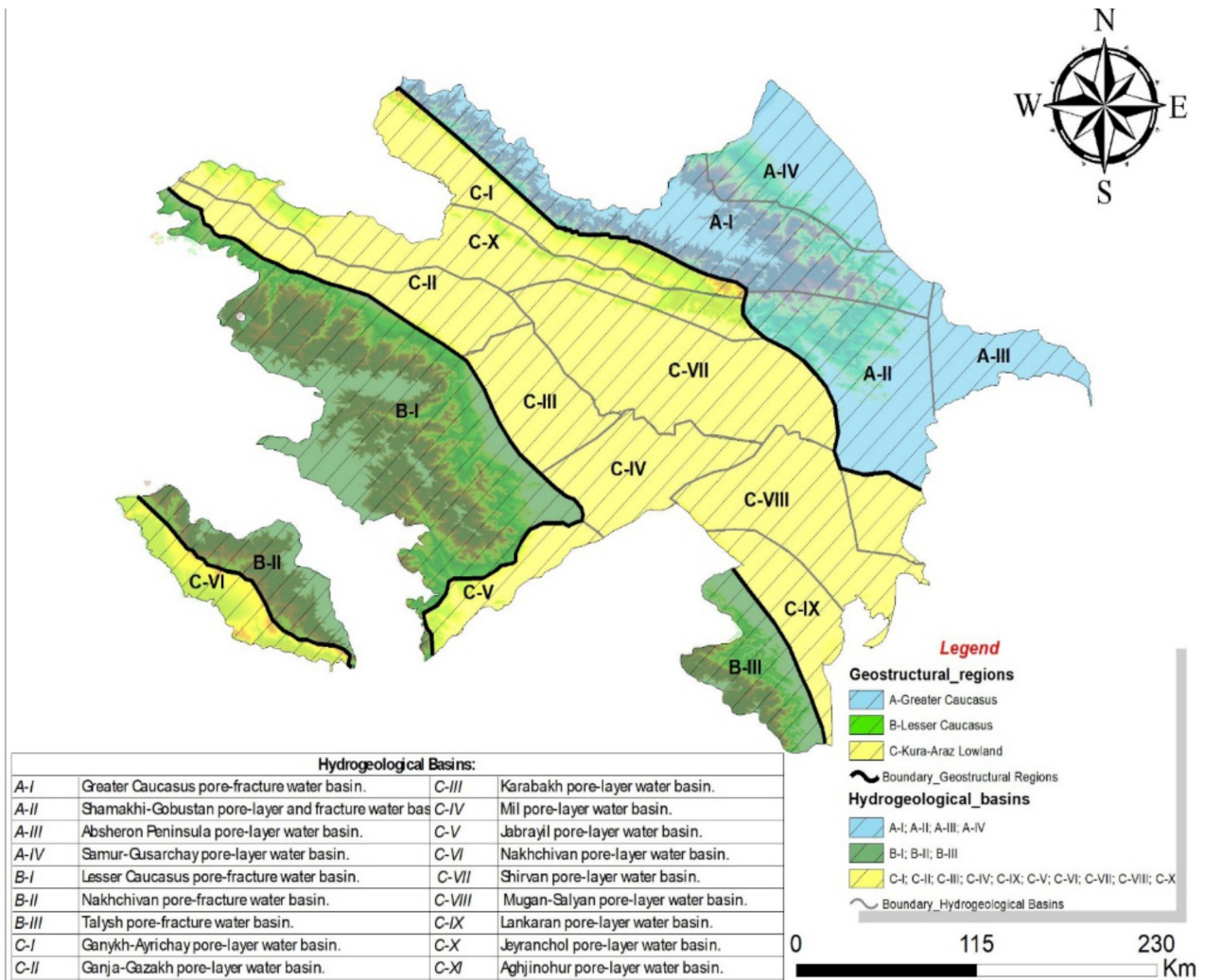


Figure 1. The Hydrogeological Basins of Azerbaijan
(İmanov and Alakbarov, 2017).

Springs are categorized based on the constancy of their flow rate as stable, low-variable, and variable; based on the amount of flow, they can be classified as low-flow (≤ 1 dm³/day), medium-flow (1-10 dm³/day), and high-flow (> 10 dm³/day). Regarding their operational duration, springs are classified into permanent, intermittent, and temporary groups (Caine & White, 1999; Böhlke & Stonestrom, 2003).

In terms of temperature, springs are generally divided into cold ($\leq 20^\circ\text{C}$), warm ($20-35^\circ\text{C}$), hot ($35-42^\circ\text{C}$), and very hot ($> 42^\circ\text{C}$) groups. Based on water mineralization levels, springs can be classified as fresh, brackish, salty, etc. Springs can also be classified according to many physical, chemical, and other properties. They are likely found in most countries around the world and play an important role in water supply, health restoration, and other fields. Sometimes, due to unsustainable human activities, the flow rate of springs decreases, water quality worsens, or a spring may completely dry up.

The study of springs has been conducted separately for different regions, considering the specific geographical, climatic, and geological characteristics of each area. Research has focused on the conditions that lead

to the formation of these springs, as well as the composition, nature, and beneficial properties of the water. Additionally, the presence of various ecosystems and groundwater resources in each region has played a crucial role in assessing the suitability of spring water for different uses. This comprehensive approach helps to determine the water quality and its potential applications across diverse regions. Depending on their hydrogeological, hydrochemical, and hydrological characteristics, springs have distinct regime features. The analysis of the discharge rates of springs located on the southern slope of the Greater Caucasus has been conducted for each administrative district. Figure 2 presents the distribution histogram of springs located on the northeastern slope of the Greater Caucasus.

The histogram clearly shows that the majority of springs with a discharge rate of less than 1 l/s are located in the Sheki district (48 springs), while the fewest are recorded in the Balakan and Gabala districts (8 springs). Based on monitoring results, springs with a discharge rate of 1-2 l/s are most commonly found in the Oguz district (11 springs), and least commonly in the Balakan district (3 springs).

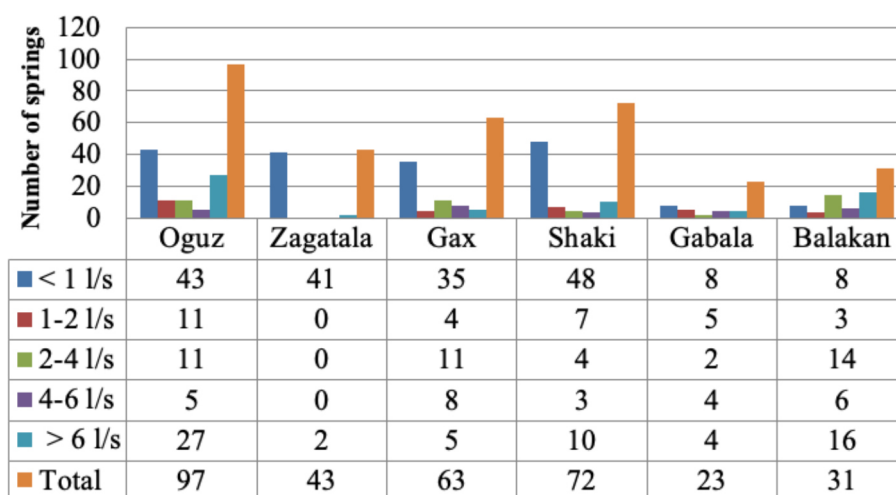


Figure 2. Distribution histogram of springs located on the southern slope of the Greater Caucasus.

In recent years, research has been conducted by the Azerbaijan State Water Resources Agency to evaluate the use of these springs as drinking water sources (Table 1). A comparison of the analysis results of samples taken from the springs in the studied area has been carried out. The main analyses included turbidity, pH, electrical conductivity, mineralization, and total hardness. As is known, according to the current "Drinking Water. Hygienic Requirements and Quality Control" AZS 929:2023 state standard in the Republic of Azerbaijan, the norm for turbidity in drinking water is 1.5 mg/l, the pH value should

be between 6 and 9, mineralization should not exceed 1000 mg/l, and total hardness should be below 7 mg-equiv/l. The water quality in all the studied springs has complied with the current standard (İsmayilov & Kreame,2026).

Based on the results of the monitoring conducted in the area, the absolute elevations at which springs are located, as well as the ranges of spring discharge and mineralization, have been provided for each administrative district of the northeastern slope of the Greater Caucasus (Table 1).

Table 1. Hydrogeological Characteristics and Spatial Variability of Springs on the Northeastern Slope of the Greater Caucasus, Azerbaijan

Administrative district	Elevation range (m a.s.l.)	Discharge range (L/s)	Mineralization range (g/L)	Hydrogeological interpretation
Gusar	770–4,090	0.10–6.00	0.10–1.00	Predominantly high-altitude springs associated with fractured and karstified mountain formations; generally low to moderate mineralization
Guba	630–2,200	0.04–7.10	0.20–1.50	Springs occurring across mountainous and foothill zones, with a relatively wide discharge variability
Shabran	23.2–605	0.02–2.50	0.30–6.60	Predominantly low-elevation springs; higher mineralization indicates stronger interaction with mineral-bearing geological formations and/or longer groundwater residence times
Siyazan	90–680	0.05–1.40	0.50–2.60	Low- to medium-elevation springs characterized by relatively moderate to high mineralization
Khizi	600–1,445	0.02–5.00	0.20–2.30	Mountain and foothill springs with considerable discharge variability and generally low to moderate mineralization

Note: Elevation is expressed in metres above sea level (m a.s.l.); discharge is expressed in litres per second (L/s); and mineralization represents total dissolved mineral content, expressed in grams per litre (g/L). The hydrogeological interpretation is based on the observed ranges and should be verified against lithological, structural, and hydrochemical data.

The analysis revealed that the highest spring discharge is observed in the Quba district (7.1 l/s), while the lowest discharge is found in the Shabran and Khizi districts (0.02 l/s). The spring with the highest mineralization is located in the Shabran district (6.6 g/l), while the spring with the lowest mineralization is situated in the Gusar district. The analysis revealed that most of the springs in the area belong

to the low and medium discharge categories. Additionally, the hydrogeological safety of the springs located in each administrative district was individually examined.

In order to determine the genesis of spring formation in the studied area, the relationship between the spring discharge and the absolute elevations, as well as between discharge and mineralization, was analyzed for

each administrative district. The analysis of figure 3 showed that in the Gusar district, as the absolute elevation increases, the spring discharge also increases. From the constructed graph's analytical expression ($Q_{av} = 0.0008H_{av}$

$- 0.3174$), it is possible to estimate the discharge of springs that were not subject to observational studies.

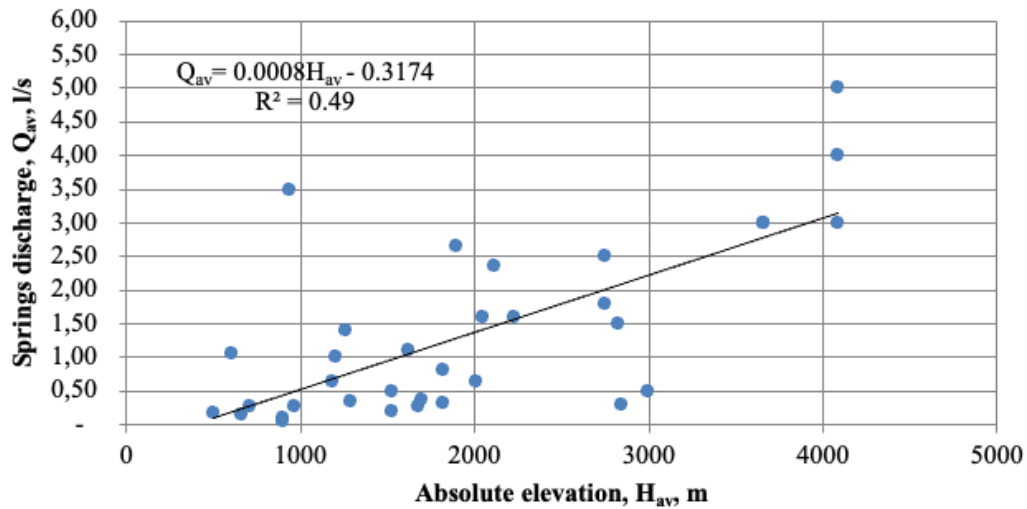


Figure 3. The relationship between the discharge of springs (Q_{av}) and the absolute elevation (H_{av}) in the Gusar region.

The graph in Figure 2 shows the relationship between spring discharge (Q_{av}) and absolute elevation (H_{av}) in the Gusar district. This graph illustrates how the discharge of springs changes with increasing elevation and can be analyzed as follows:

Analysis of the Graph:

- X-axis: Absolute Elevation (H_{av}) — The elevation of the spring's location above sea level, measured in meters.
- Y-axis: Spring Discharge (Q_{av}) — The amount of water flowing from the spring, measured in liters per second (l/s).

Analytical Description of the Graph:

The analytical expression for the graph is:
 $Q_{av} = 0.0008 \times H_{av} - 0.3174$

This formula helps to determine the relationship between absolute elevation (H_{av}) and spring discharge (Q_{av}). The analysis shows that:

1. As the absolute elevation increases, the spring discharge also increases. The graph indicates a positive correlation between elevation and discharge. This suggests that higher elevations are likely to have springs with greater flow rates.

2. The analytical expression allows for estimating the discharge of springs based on their elevation. For example, to calculate the discharge of a spring located at 1500 meters elevation:
 $Q_{av} = 0.0008 \times 1500 - 0.3174 = 1.1826$ l/s

This graph and the analytical expression clarify the relationship between spring discharge and absolute elevation in the Gusar district, providing valuable information for hydrogeological studies and enabling more accurate assessments based on this data. The same analysis has been conducted for all districts.

In general, despite the weak distribution of alluvial-deluvial deposits in the Greater Caucasus, groundwater is sporadically found in these rocks. Most of the springs associated with alluvial-deluvial rocks have a discharge of less than 1 l/s. However, there are also springs with discharges reaching up to 25 l/s. The degree of

mineralization of spring waters in alluvial-deluvial deposits ranges from 0.15 to 0.25 g/l, and their chemical composition is predominantly calcium bicarbonate and calcium-magnesium bicarbonate, with some cases exhibiting more complex compositions. In the glacial deposits widespread on the peaks of Bazarduzu and Shahdag, perennial springs with descending flow rates ranging from 0.1 to 30.0 l/s have developed. The mineralization degree of the water from these springs ranges from 0.2 to 0.4 g/l, and their chemical composition is primarily calcium bicarbonate (Geology of Azerbaijan, 1961; İmanov & Alakbarov, 2017; İsmayilov, 2019).

In the mountainous regions, the groundwater of mountain rivers is of significant importance for practical use. The discharge of springs fed by the groundwater of these rivers varies over a wide range. For instance, the discharge of a spring near the village of Yerfi on the Babachay riverbed is 23 l/s, while the discharge of a spring near the village of Valasli on the Goychay riverbed reaches 30 l/s. The mineralization of the water from both springs is up to 1 g/l, and their chemical composition is calcium bicarbonate.

In the part of the Lesser Caucasus mountain-fold zone located in Azerbaijan and in the Mountainous Talysh, numerous springs are related to all stratigraphic units from the Quaternary to Carboniferous and Devonian. In these areas, fractured and, in some places, karstified limestone rocks are more water-bearing, while volcanic rocks have relatively low water content. Intrusive formations, in particular, are characterized by very poor water yield. In the volcanogenic-tuff rocks of the Lesser Caucasus, numerous springs with discharges ranging from 8-10 l/s, and sometimes reaching 50-70 l/s, are manifested. The mineralization degree of the calcium bicarbonate waters of the volcanogenic rocks typically does not exceed 0.1-0.5 g/l.

According to the geostructural regional characteristics, the territory of the Lankaran natural region is divided into the Lesser Caucasus mountain-fold system

and the Kura-Araz lowland. The springs located in the area are found in both the Talish fracture-porous water and Lankaran porous-layer water basins. In Talish, the discharge of springs associated with various rocks typically ranges from 0.1-1.0 l/s to 2-3 l/s, and in some cases, it reaches up to 10 l/s. The Lankaran foothill plain, which holds exceptional importance in the hydrogeology of Azerbaijan, is a region rich in potable and low-salinity groundwater within the porous-layer water basin.

The analysis revealed that most of the springs in the area belong to the low discharge spring class.

Additionally, the hydrogeological safety of the springs located in each administrative district has been individually examined. It was determined that the quality of most of the springs in the area meets the standards. An analysis of the discharge of springs in the Lankaran natural region was conducted according to the number of springs in each administrative district. Figure 4 shows the distribution histogram of the springs located in the Lankaran natural region.

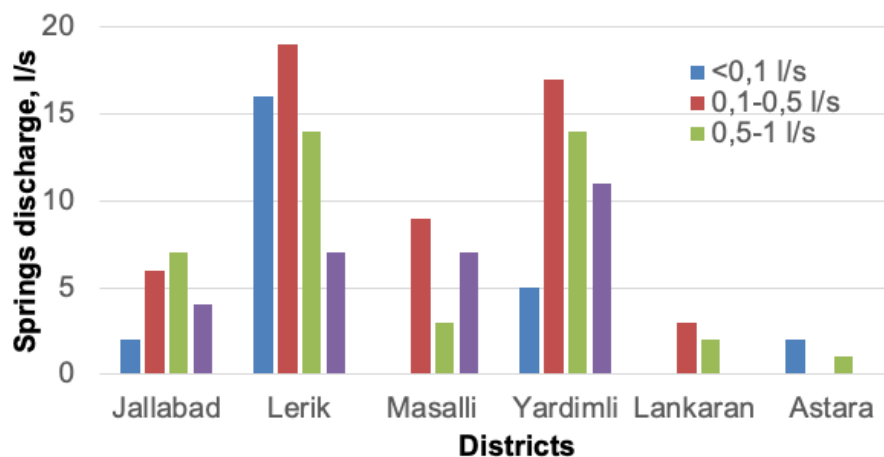


Figure 4. *Distribution Histogram of Springs Located in the Lankaran Natural Region.*

The histogram presented in Figure 4 illustrates the frequency distribution of springs in the Lankaran natural region according to their discharge rates. The distribution provides an indication of the relative prevalence of springs across different discharge classes and thereby contributes to the characterization of the regional spring-flow regime. The observed distribution suggests that spring discharge is heterogeneous, reflecting differences in groundwater recharge, aquifer hydraulic properties, lithological composition, topographic setting, and groundwater flow-path characteristics.

The hydrogeological assessment further indicates a spatial differentiation in spring-water quality across the Lankaran natural region. Springs located within higher-elevation catchment areas generally exhibit comparatively favorable physicochemical characteristics. This pattern may be associated with differences in recharge conditions, land-use intensity, groundwater residence time, and the degree of anthropogenic influence. In mountainous and relatively undisturbed recharge zones, precipitation-derived recharge may reach groundwater systems more rapidly and with comparatively limited exposure to anthropogenic sources of contamination. However, elevation itself should not be considered an independent determinant of water quality, as spring-water chemistry is governed by the combined effects of lithology, water-rock interaction, groundwater residence time, recharge processes, and land use (Böhlke & Stonestrom, 2003; Alakbarov, 2014; İsmayilov & Kreame, 2026).

The assessment of spring-water quality by administrative district reveals a generally favorable hydrochemical and physicochemical status for a substantial proportion of the investigated springs, although several

local exceptions were identified. The spatial variability in water quality is likely related to the contrasting geological and hydrogeological conditions of individual catchments, as well as differences in settlement density, agricultural activity, land-use patterns, and potential pollution pressures. Consequently, the assessment of spring quality should be considered within the context of the specific hydrogeological setting and recharge characteristics of each spring rather than solely on the basis of its administrative location.

A pronounced spatial concentration of springs was observed in the Lerik and Yardimli districts, which contain a substantial proportion of the springs documented within the Lankaran natural region (İsmayilov & Kreame, 2026). The relatively high density of springs in these mountainous districts can be associated with their complex relief, substantial precipitation and recharge potential, and the presence of fractured and permeable geological formations that facilitate groundwater circulation and discharge. The interaction between topography, geological structure, and groundwater flow systems creates favorable conditions for the emergence of springs along lithological contacts, fracture zones, and other hydrogeological discontinuities.

The predominance of springs in the mountainous sectors of the Lankaran region is also of particular significance from a water-resources and ecological perspective. Springs provide localized groundwater discharge that contributes to headwater streams and supports aquatic and terrestrial ecosystems, particularly during periods of reduced surface-water availability. Their spatial distribution and discharge characteristics can therefore serve as important indicators of groundwater-

system functioning and catchment-scale hydrological processes (Barber & Wheeler, 2005; Caine & White, 1999).

Overall, the results demonstrate that the spring systems of the Lankaran natural region exhibit considerable spatial variability in both discharge and water quality. The concentration of springs in Lerik and Yardimli, together with the comparatively favorable quality observed in many higher-elevation catchments, highlights the importance of protecting mountainous recharge areas from land-use pressures and potential contamination sources. Long-term monitoring of spring discharge and hydrochemical characteristics, combined with systematic assessment of recharge zones and groundwater–surface-water interactions, is therefore essential for sustainable utilization and conservation of these groundwater resources (İmanov & Alakbarov, 2017; İsmayilov et al., 2026).

The hydrogeological conditions of the Nakhchivan Autonomous Republic (Nakhchivan AR) are characterized by complexity and diversity, shaped under the influence of a wide range of natural and anthropogenic factors. In Nakhchivan AR, numerous springs are related to all stratigraphic units from the Quaternary to Carboniferous and Devonian periods. Fractured and, in some areas,

karstified limestone rocks are more water-bearing, while volcanic rocks are less water-bearing, and intrusive formations, in particular, have very poor water yield. The studied area has created a highly complex pressurized water system from a hydrogeological perspective. Here, the flow of groundwater is observed through fractures with different genesis, and in some places (on slopes, in river valleys, etc.), it is found in eroded sedimentary rocks. Groundwater recharge primarily occurs through atmospheric precipitation and condensation water, and its discharge is formed by draining into the hydrographic network and the spring system. In the area, depending on the elevation levels of the relief, the number of springs, the intensity of their recharge, the vertical zonality of hydrogeological parameters, and the decrease in the mineralization of spring water with increasing elevation are observed. Additionally, high endogenous heat flow is present, which results in the natural discharge of thermal waters along tectonic fractures. Based on the results of the monitoring conducted in the area, the absolute elevations, discharge, and mineralization ranges of springs for each administrative district of the Nakhchivan Autonomous Republic (Nakhchivan AR) are provided in Table 2.

Table 1. *Hydrogeological and Hydrochemical Characteristics of Springs along the Northeastern Slope of the Greater Caucasus, Azerbaijan*

Administrative district	Spring elevation range (m a.s.l.)	Spring discharge range (L s ⁻¹)	Mineralization range (g L ⁻¹)
Gusar	770–4,090	0.10–6.00	0.10–1.00
Guba	630–2,200	0.04–7.10	0.20–1.50
Shabran	23.2–605	0.02–2.50	0.30–6.60
Siyazan	90–680	0.05–1.40	0.50–2.60
Khizi	600–1,445	0.02–5.00	0.20–2.30

Note: Spring elevation is given in metres above sea level (m a.s.l.); discharge is expressed in litres per second (L s⁻¹); and mineralization represents the total dissolved mineral content, expressed in grams per litre (g L⁻¹). The reported values represent the observed ranges within each administrative district.

The investigated springs of the Nakhchivan Autonomous Republic exhibit pronounced spatial variability in elevation, discharge, and mineralization, reflecting the heterogeneous geological, geomorphological, and hydrogeological conditions of the region (Table 2). Spring elevations range from 790 to 3,065 m a.s.l., with the highest-elevation springs occurring in the Shahbuz District (1,090–3,065 m a.s.l.). In comparison, springs in Babek (790–1,380 m a.s.l.), Julfa (950–1,070 m a.s.l.), Ordubad (800–1,390 m a.s.l.), and Sharur (840–1,270 m a.s.l.) occupy considerably narrower elevation ranges. The extensive altitudinal distribution of springs in Shahbuz is consistent with its mountainous topography and complex geological setting, where variations in lithology, tectonic structure, fracture development, and recharge conditions exert important controls on groundwater circulation and discharge (Geology of Azerbaijan, 1961; Geology of Azerbaijan, 2008; Geology of Azerbaijan, 2015).

Spring discharge demonstrates substantial spatial heterogeneity across the investigated districts. The maximum observed discharge occurs in Sharur, where values range from 0.40 to 139 L s⁻¹, while Shahbuz exhibits

a discharge range of 0.10–51.0 L s⁻¹. Considerably narrower discharge ranges are observed in Babek (0.50–8.00 L s⁻¹), Julfa (1.00–7.00 L s⁻¹), and Ordubad (5.00–10.00 L s⁻¹). Such variability reflects differences in aquifer transmissivity, permeability, fracture connectivity, recharge magnitude, hydraulic gradients, and groundwater flow-path characteristics (Barber & Wheeler, 2005; Caine & White, 1999; Alakbarov, 2014). In particular, high-discharge springs may represent preferential groundwater-flow pathways associated with highly fractured, faulted, or karstified formations, whereas low-discharge springs may occur where groundwater movement is constrained by less permeable geological units or limited recharge (Caine & White, 1999; Böhlke & Stonestrom, 2003).

Spring-water mineralization ranges from 0.10 to 3.70 g L⁻¹ across the Nakhchivan districts. The widest mineralization range is recorded in Shahbuz (0.10–3.70 g L⁻¹), whereas Ordubad exhibits the narrowest range (0.20–0.50 g L⁻¹). Sharur also demonstrates comparatively elevated mineralization, with values ranging from 0.30 to 1.80 g L⁻¹. Variations in mineralization are commonly associated with differences in lithological composition,

groundwater residence time, water–rock interaction, flow-path length, and mixing processes within the aquifer system (Böhlke & Stonestrom, 2003; Alakbarov, 2000). Consequently, mineralization can provide a useful first-order indicator of groundwater evolution and hydrochemical differentiation, although it should not be interpreted as a direct measure of potable water quality without complementary physicochemical and microbiological analyses.

Nakhchivan is also characterized by a high diversity and concentration of mineral springs. More than 200 mineral springs have been reported within the autonomous republic, with many occurring in the valleys and catchments of the Eastern Arpaçay, Nakhchivançay, Alinjaçay, Gilançay, Ordubadçay, Qaradərə, and Ayılış rivers. Their spatial distribution is closely related to the geological and structural framework of the region, including fractured rocks, tectonic discontinuities, lithological contacts, and formations of different geological ages (Geology of Azerbaijan, 1961; Geology of Azerbaijan, 2008; Geology of Azerbaijan, 2015). These geological characteristics provide preferential pathways for groundwater infiltration, storage, circulation, and discharge and therefore play a fundamental role in determining the location and hydrochemical characteristics of springs (Alakbarov, 2014).

The hydrochemical composition of mineral springs in Nakhchivan is characterized by the presence of major cations, including Ca^{2+} , Mg^{2+} , and Na^+ K^+ , and major anions such as HCO_3^- , SO_4^{2-} , and Cl^- . The relative contribution of these ions varies according to lithology, groundwater residence time, mineral dissolution, ion exchange, and other water–rock interaction processes occurring along groundwater flow paths (Böhlke & Stonestrom, 2003; Alakbarov, 2000). Such hydrochemical variability highlights the importance of considering individual groundwater-flow systems rather than treating mineral springs as a chemically homogeneous resource.

The hydroecological significance of Azerbaijan's spring systems extends beyond their role as natural groundwater discharge points. Springs form important hydrological interfaces between groundwater and surface-water systems and can sustain headwater streams, wetlands, riparian habitats, and other groundwater-dependent ecosystems. The magnitude and temporal stability of spring discharge are therefore important components of catchment-scale hydrological functioning, particularly during periods of reduced precipitation and low surface-water availability (Barber & Wheeler, 2005; İsmayilov & Kreame, 2026). Changes in groundwater recharge, abstraction, land-use patterns, and climatic conditions may modify both the quantity and quality of spring discharge, potentially affecting spring-dependent ecosystems and downstream water availability (İmanov & Alakbarov, 2017; İsmayilov et al., 2026).

At the national scale, the distribution and hydrochemical characteristics of springs are strongly controlled by the contrasting geological and geomorphological settings of the Greater Caucasus, Lesser Caucasus, and Kura–Araz Lowland. Fractured and karstified carbonate formations may facilitate rapid groundwater circulation through preferential pathways, whereas volcanic, volcanoclastic, and less permeable

sedimentary formations can impose different controls on groundwater storage and transmission (Geology of Azerbaijan, 2008; Geology of Azerbaijan, 2015). Tectonic faults, fracture zones, erosion surfaces, and lithological contacts may consequently act as important structural controls on groundwater movement and spring emergence. Depending on the local hydraulic configuration, these conditions may give rise to both descending and ascending springs (Alakbarov, 2014).

The relationship between elevation, discharge, and mineralization should therefore be interpreted within the broader context of groundwater-system architecture. Although springs at higher elevations may in some locations exhibit relatively low mineralization, this pattern should not be generalized as a direct relationship between elevation and water quality. Groundwater chemistry is controlled by a combination of recharge characteristics, lithological composition, groundwater residence time, flow-path length, water–rock interaction, and anthropogenic pressures (Böhlke & Stonestrom, 2003; Alakbarov, 2014). Similarly, spring discharge cannot be attributed solely to elevation because aquifer properties, recharge rates, hydraulic gradients, structural permeability, and catchment characteristics may exert stronger controls on spring yield (Barber & Wheeler, 2005; Caine & White, 1999).

The hydrogeological assessment of springs intended for drinking-water use should consequently incorporate a comprehensive suite of physicochemical, hydrochemical, and microbiological parameters. In addition to mineralization, parameters such as pH, electrical conductivity, turbidity, hardness, major and trace elements, and microbiological indicators should be systematically evaluated. Previous investigations of springs on the northeastern slope of the Greater Caucasus have emphasized the importance of assessing spring-water quality against drinking-water requirements (İsmayilov, 2019). Such integrated monitoring is essential for distinguishing naturally mineralized groundwater from water affected by anthropogenic contamination and for determining the suitability of individual springs for domestic or drinking-water supply.

From a water-resources management perspective, the pronounced spatial and hydrochemical variability of springs indicates that their assessment and utilization should be based on site-specific hydrogeological investigations. Sustainable management requires the protection of groundwater recharge zones, regulation of abstraction, monitoring of spring discharge and water chemistry, and identification of potential contamination sources. Climate change may further modify groundwater recharge through changes in precipitation patterns, evapotranspiration, drought frequency, and the seasonal distribution of water availability, thereby increasing the need for adaptive groundwater-management strategies (İmanov & Alakbarov, 2017; İsmayilov et al., 2026).

Overall, the findings demonstrate that Azerbaijan's springs constitute highly heterogeneous groundwater systems controlled by the interaction of geological structure, lithology, geomorphology, recharge processes, groundwater circulation, and anthropogenic influences. The observed variability in spring elevation, discharge, and mineralization, particularly in the

Nakhchivan Autonomous Republic, highlights the need for an integrated hydrogeological and hydroecological assessment framework. Combining spring inventories with discharge monitoring, hydrochemical characterization, geological and structural analysis, and ecological assessment can provide a scientifically robust basis for evaluating groundwater resources, identifying vulnerable spring systems, protecting groundwater-dependent ecosystems, and developing sustainable spring-water management strategies under increasing climatic and anthropogenic pressures (Alakbarov, 2014; Barber & Wheeler, 2005; Böhlke & Stonestrom, 2003; İsmayilov & Kreame, 2026).

4. Conclusion

The hydrogeological study of springs in Azerbaijan reveals a complex and diverse water system shaped by the region's varied geological and climatic conditions. Springs in the Greater Caucasus, Lesser Caucasus, Talish, and Kura-Araz Lowland areas exhibit distinct hydrological characteristics, influenced by factors such as lithology, tectonic structures, and erosion zones. The classification of springs based on discharge rates, mineralization, and temperature provides a comprehensive understanding of their potential for water supply, with many springs offering low to medium flow rates and mineralization levels that make them suitable for various uses, including drinking water.

Geological formations, including fractured, karstified, and porous layers, significantly affect the quantity and quality of groundwater, contributing to the formation of both descending and ascending springs. These natural water sources are vital for local communities and ecosystems, offering essential water resources in a region where groundwater recharge is heavily reliant on precipitation.

Hydrogeological investigations are critical in ensuring that spring water meets the required standards for human consumption. The analysis of physical and chemical parameters such as turbidity, pH, electrical conductivity, and mineralization confirms that most of the springs in the studied areas are suitable for drinking, provided that they adhere to the national water quality standards. However, challenges remain, particularly in areas with higher mineralization, where careful monitoring is required to maintain water quality and prevent contamination.

The study also highlights the importance of sustainable water management practices, as unsustainable exploitation of groundwater resources or contamination can lead to declines in both water quantity and quality. The balance between groundwater recharge and extraction must be maintained to ensure the long-term viability of spring-fed water supplies.

In summary, the hydrogeological characteristics of Azerbaijan's springs offer valuable insights into the region's water resources. These springs, influenced by diverse geological features and hydrological conditions, represent a critical component of Azerbaijan's water supply system. Ongoing research and monitoring are essential to preserve these valuable resources, ensuring they remain safe and reliable for future generations.

CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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